

Powering the Sun

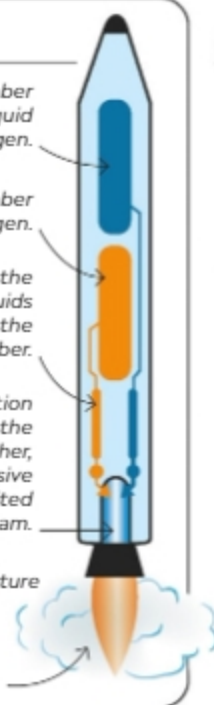
The sun is a massive ball of flaming hydrogen and helium. The source of the sun's energy is its hot, dense core, where hydrogen atoms undergo a process of fusion to produce the gas helium. More than 661 tons of hydrogen are converted into helium each second, releasing intense heat and light.

The sun is about 75 percent hydrogen.

ROCKET FUEL

Some space rockets are powered by liquid hydrogen. This element combines with liquid oxygen to create extremely hot steam, which escapes out of the nozzle with great force. This produces a thrust that pushes the rocket upward. As an added benefit, this process generates only steam as a by-product, making hydrogen an environmentally safe choice of fuel.

1. This chamber contains liquid hydrogen.
2. This chamber contains liquid oxygen.
3. Pumps control the flow of the liquids as they enter the combustion chamber.
4. The combustion chamber is where the liquids come together, creating an explosive mixture that is ignited to create hot steam.
5. The explosive mixture passes through the nozzle, creating steam that pushes the rocket upward.



Hydrogen-powered bus in Japan

Clean fuel

Pure hydrogen is a clean energy source and is used to power some vehicles. When hydrogen is passed through a fuel cell, it chemically fuses with oxygen from the air. This reaction produces electricity, which powers the motor inside the vehicle.



Fritz Haber on a German postage stamp

Making ammonia

The combination of hydrogen and nitrogen produces ammonia, which is used to make fertilizers that aid plant growth. Known as the Haber Process, this technique was developed by the German chemist Fritz Haber in the early 20th century and is still widely used.

Hindenburg disaster

In 1936, the German airship *Hindenburg* was the biggest aircraft ever built. It was filled with hydrogen, which easily catches fire and burns with an extremely hot flame. In 1937, the *Hindenburg* burned when the hydrogen gas began to leak and caught fire. The accident killed 36 people.

